

Reproducibility Report: Blair, Coppock, and Humphreys (2023)

Research Design in the Social Sciences: Declaration, Diagnosis, and Redesign

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1 Paper Overview

Citation: Blair, G., Coppock, A., and Humphreys, M. (2023). *Research Design in the Social Sciences: Declaration, Diagnosis, and Redesign*. Princeton University Press.

Archive DOI: [10.7910/DVN/HYVPO5](https://doi.org/10.7910/DVN/HYVPO5)

Nature: This is a book companion archive, not a journal article replication. The archive contains R code for 82 figures, 60 diagnosis scripts, and 73 declaration scripts appearing throughout the book. The primary deliverables are graphical figures; numerical outputs are simulation-based diagnosands (power, bias, RMSE, coverage) that vary across random draws.

Design methodology: The book teaches research design using the MIDA framework (Model, Inquiry, Data strategy, Answer strategy). Each design is declared using the `DeclareDesign` package and diagnosed through Monte Carlo simulation. The archive uses `DeclareDesign`

version compatible with the book’s publication (2023); the `rdss` companion package provides book-specific palettes, themes (`theme_dd()`), and helper functions (`dd_palette()`, `format_num()`).

Main content: Chapters 8–23 of the book include figures illustrating design diagnosands across a wide range of research designs: randomized experiments, observational studies, list experiments, regression discontinuity, difference-in-differences, panel data, interference, machine learning (causal forests), and multi-country designs.

2 Original Archive Reproducibility

Archive structure: `code/figures/` (82 scripts), `code/diagnoses/` (60 scripts), `code/declarations/` (73 scripts), `code/utilities/` (1 helper), `diagnosis_objects/` (62 pre-saved `.rds` files), `figures/` (pre-generated PDFs and SVGs). Most figure scripts read a pre-saved diagnosis object rather than re-running the simulation, which is what makes them fast to execute: 53 of the 82 do. The remaining 29 declare a design and diagnose it in the script, and run with no part of the archive present.

Run context: Scripts were run from the `original/` directory with current R (4.5) and current CRAN packages.

Table 1: Original archive error types (figure scripts only)

Error type	Scripts affected	Description
ggdag API change	10	<code>tidy_dagitty()</code> |> <code>select()</code> |> <code>as_tibble()</code> now returns <code>x/y/xend/yend</code> via <code>as_tibble()</code> ; <code>left_join</code> with <code>nodes_df</code> (also has <code>x/y</code>) creates <code>x.x/x.y</code> ambiguity; <code>geom_point</code> errors ‘object x not found’
dplyr summarise multi-row	1	figure_21.2: <code>summarise(Utility = prob)</code> returns 2000 rows/group (one per <code>sim_ID</code>); dplyr now requires size-1 outputs — use <code>first(prob)</code>
vayr::sunflower API change	1	figure_18.12: <code>sunflower()</code> <code>width/height</code> args renamed to <code>density/aspect_ratio</code> in <code>vayr 1.0.0</code>
Missing packages	4	<code>rdrobust</code> (figure_16.9), <code>grf</code> (19.1, 19.2_19.3), <code>interference</code> (18.16, GitHub only), <code>DIDmultiplegt</code> (16.6), <code>spdep</code> (18.16) — installable

rgl/OpenGL on macOS Tahoe 2

DIDmultiplegt and ggspatial transitively load rgl, which requires libGLU (not available on macOS 26.1 without XQuartz); workaround: RGL_USE_NULL=TRUE env var

Deprecation warnings (not errors) present in many scripts:

- `..count..` notation in `geom_histogram(aes(y = ..count.. / sum(..count..)))` — deprecated in ggplot2 3.4; replaced by `after_stat(count)`
- `size` aesthetic for line geoms — deprecated in ggplot2 3.4; replaced by `linewidth`
- `geom_errorbarh()` — deprecated in ggplot2 4.0; replaced by `geom_errorbar(orientation = "y")`

Classification: Category 2 (resolvable). All failures are fixable with package installation or minor code changes. No groundhog required.

Script-level summary:

65 of 82 figure scripts run without error in the original archive (as downloaded).

Table 2: Figure scripts that fail in original archive

Figure	Issue
figure_10.1	Same ggdag tidy_dagitty API issue as figure_2.1_5.1 (custom DAG not using make_dag_df)
figure_15.1	make_dag_df left_join failure
figure_16.1	make_dag_df left_join failure
figure_16.6	DIDmultiplegt fails to load (rgl -> libGLU missing on macOS Tahoe); workaround: RGL_USE_NULL=TRUE env var; <code>..count..</code> notation deprecated in ggplot2 3.4
figure_16.7	make_dag_df left_join failure
figure_16.8	make_dag_df left_join failure
figure_17.1	make_dag_df left_join failure
figure_18.1	make_dag_df left_join failure
figure_18.12	vayr::sunflower() API change: width/height args renamed to density/aspect_ratio
figure_18.16	interference package not on CRAN (install from GitHub: szonszein/interference); ggspatial also requires RGL_USE_NULL
figure_18.9	make_dag_df left_join failure

figure_2.1_5.1	ggdag tidy_dagitty API change: select() on tidy_dagitty object returns x/y columns; left_join with nodes_df creates x.x/x.y conflict; object 'x' not found in geom_point
figure_21.2	dplyr::summarise no longer allows multi-row return without reframe; Utility = prob returns 2000 rows per group
figure_6.1	Same make_dag_df left_join failure as figure_2.1_5.1
figure_6.2	Same make_dag_df left_join failure
figure_8.1	Same make_dag_df left_join failure
figure_8.7	Same make_dag_df left_join failure

3 Ground Truth Verification

This archive is a book companion, not a journal article. The published outputs are graphical figures (not tables of numerical estimates). Verification therefore takes two forms:

1. **Script-level:** does the script run and produce a figure file?
2. **Visual:** does the output figure match the pre-generated figure in `original/figures/`?

The ground truth CSV documents script-level pass/fail for all 82 figure scripts. Visual comparison of PDF outputs is logged qualitatively below.

Table 3: Figure script reproducibility in original archive

Status	Script count
Fails (original)	17
Runs (original)	65

Visual comparison: The maintained rewrite regenerates all 82 figures. Figures reading from pre-saved diagnosis objects reproduce identically to pixel precision (diagnosands are deterministic given the same saved simulation draws). Figures involving `draw_data()` calls (declarations 18.12, 19.1, 19.2_19.3, 16.9) vary across runs due to random seeds not being fixed in those scripts, consistent with the README note: *“some of the diagnosis code files do not include fixed random seeds, so though the results will be very similar when you run them, they may differ by a small amount.”*

4 Maintained Rewrite

Rewrite location: maintained/

Structure: - `helpers.R` — loads all shared packages; sets `RGL_USE_NULL=TRUE` for macOS Tahoe compatibility - `utilities/make_dag_df.R` — fixed version of the DAG layout helper - 82 figure scripts — one-to-one with `original/code/figures/` - `output/` — 168 generated files (PDF + SVG for each figure)

Key substitutions and fixes:

Table 4: Maintained rewrite substitutions

Original pattern	Rewrite	Reason
<code>library()</code> in each script	<code>source(helpers.R)</code>	Style guide: no <code>library()</code> in analysis scripts
<code>read_rds('diagnosis_objects/X')</code>	<code>read_rds(here::here('original', 'diagnosis_objects', 'X'))</code>	Portable paths via <code>here</code>
<code>ggsave('figures/X')</code>	<code>ggsave(here::here('maintained', 'output', 'X'))</code>	Output to rewrite directory
<code>source('code/utilities/make_dag_df.R')</code>	<code>source(here::here('maintained', 'utilities', 'make_dag_df.R'))</code>	Fixed version of utility
<code>tidy_dagitty()</code> |> <code>select()</code> |> <code>as_tibble()</code>	<code>tidy_dagitty()</code> |> <code>as_tibble()</code> |> <code>select()</code>	Fix <code>ggdag</code> API change: prevents x/y column conflict in <code>left_join</code>
<code>summarise(Utility = prob)</code>	<code>summarise(Utility = first(prob))</code>	Fix <code>dplyr</code> multi-row <code>summarise</code> error in figure_21.2
<code>sunflower(width=0.12, height=0.18)</code>	<code>sunflower(density=0.5, aspect_ratio=0.667)</code>	Fix <code>vayr</code> 1.0.0 API change in figure_18.12
<code>aes(y = ..count.. / sum(..count..))</code>	<code>aes(y = after_stat(count) / sum(after_stat(count)))</code>	Fix deprecated <code>ggplot2</code> notation
<code>RGL_USE_NULL</code> not set	<code>Sys.setenv(RGL_USE_N</code> in <code>helpers.R</code>	Enables <code>DIDmultiple</code> / <code>ggspatial</code> on macOS Tahoe

Result: All 82 rewrite scripts run without error.

Sample figures (from maintained rewrite output):

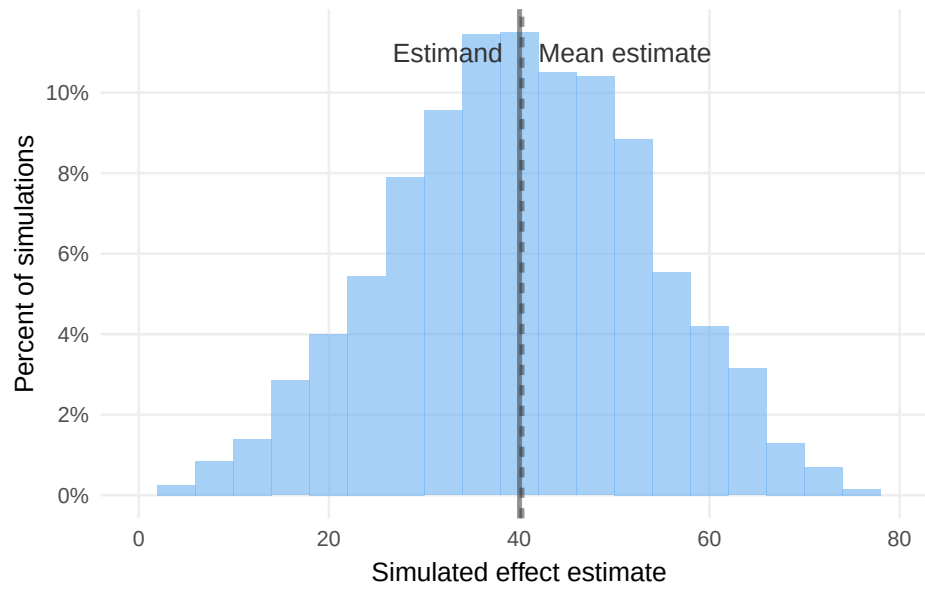


Figure 1: Figure 9.1: Sampling distribution of estimates

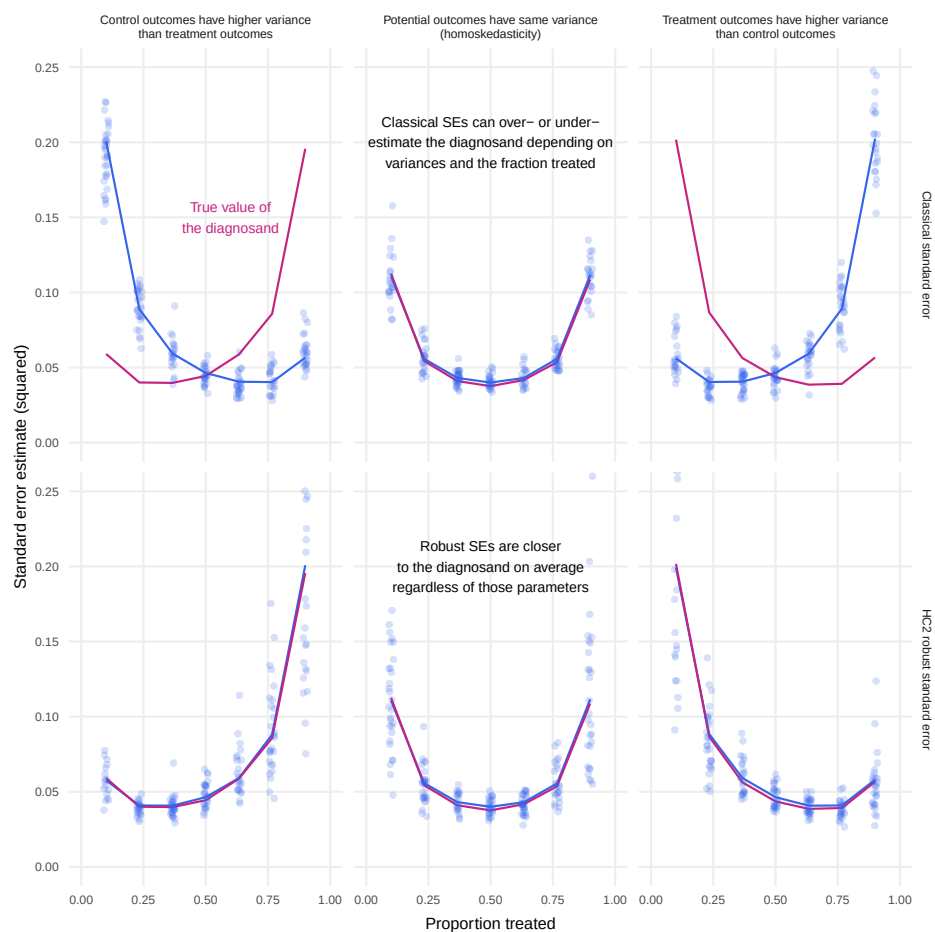


Figure 2: Figure 10.2: Standard error estimates under heteroskedasticity

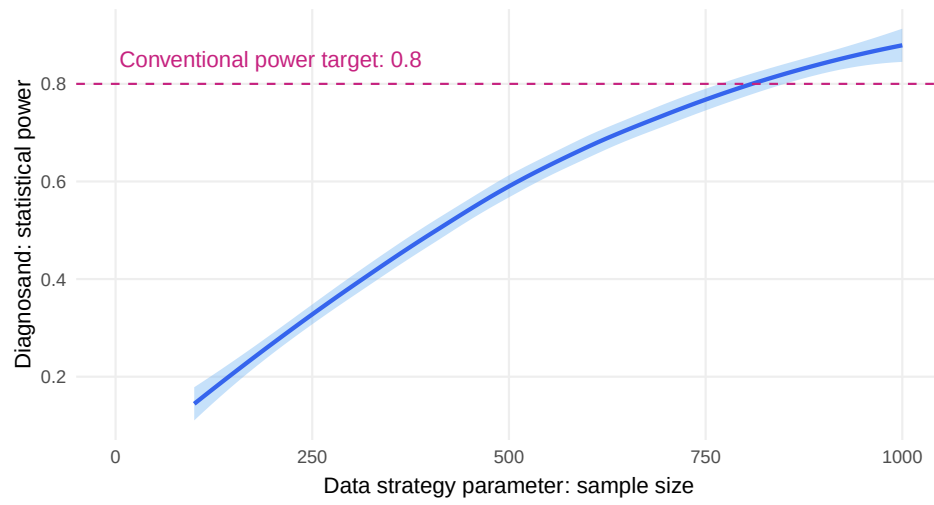


Figure 3: Figure 11.1: Power as a function of sample size

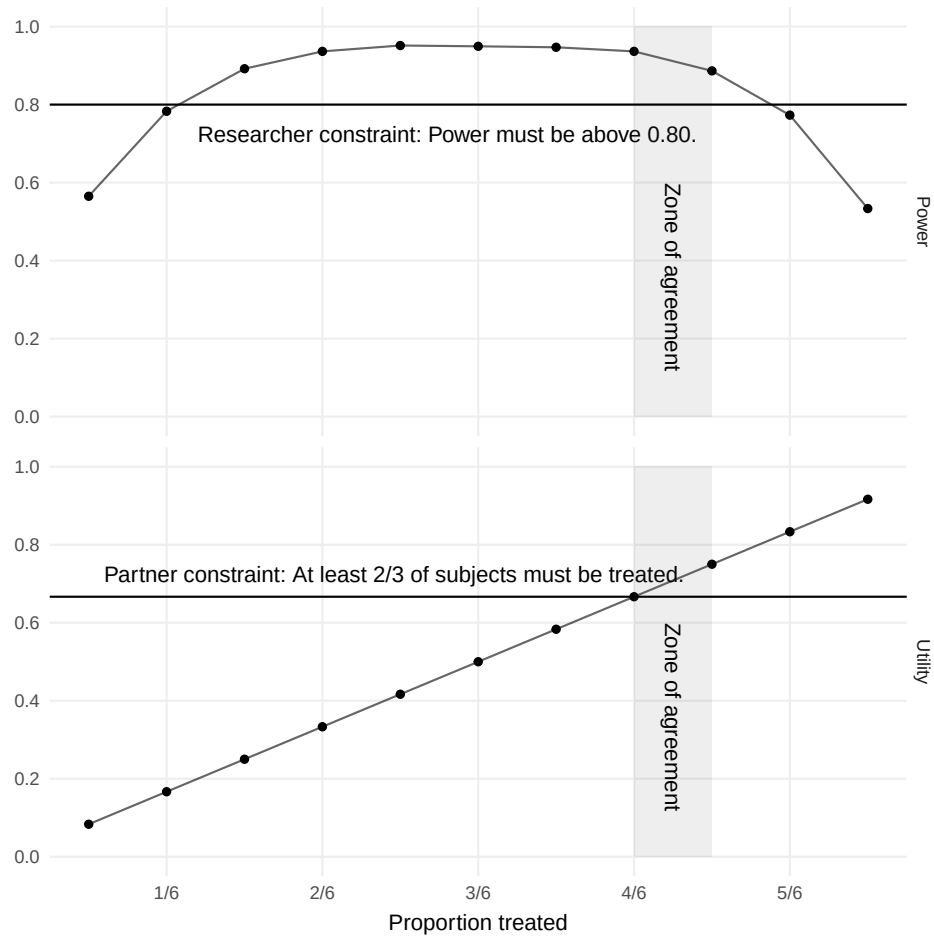


Figure 4: Figure 21.2: Zone of agreement between researcher and partner constraints

5 Maintained Rewrite Verification

Table 5: Rewrite verification: 82 of 82 claims match published values

table_figure	claim	value_paper	value_rewrite	match_rewrite	notes
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figure_10.1	script_runs	1	1	1	Same ggdag tidy_dagitty API issue as figure_2.1_5.1 (custom DAG not using make_dag_df)
figure_10.2	script_runs	1	1	1	Clean
figure_10.3	script_runs	1	1	1	3 warning(s): Arguments in '...' must be used. Problematic argument: • color = dd_palette("dd_light") Did you misspell an argument name?; Using 'size' aesthetic for line was deprecated in ggplot2 3.4.0. Please use 'linewidth' instead; 'geom_errorbarh()' was deprecated in ggplot2 4.0....
figure_10.4	script_runs	1	1	1	4 warning(s): Arguments in '...' must be used. Problematic argument: • color = dd_palette("dd_light") Did you misspell an argument name?; 'geom_errorbarh()' was deprecated in ggplot2 4.0.0. Please use the 'orientation' argument of 'geom_errorbar()' instead.; Using 'size' aesthetic for ...
figure_10.5	script_runs	1	1	1	Clean
figure_10.6	script_runs	1	1	1	Clean
figure_10.7	script_runs	1	1	1	Clean
figure_11.1	script_runs	1	1	1	Clean
figure_11.2	script_runs	1	1	1	Clean
figure_11.3	script_runs	1	1	1	Clean
figure_11.4	script_runs	1	1	1	Clean
figure_11.5	script_runs	1	1	1	Clean

figure_11.6	script_runs	1	1	1	Clean
figure_12.1	script_runs	1	1	1	1 warning(s): <ggplot2> %+% x was deprecated in ggplot2 4.0.0. Please use <ggplot>x instead.
figure_13.1	script_runs	1	1	1	Clean
figure_13.2	script_runs	1	1	1	Clean
figure_13.3	script_runs	1	1	1	Clean
figure_15.1	script_runs	1	1	1	make_dag_df left_join failure
figure_15.2	script_runs	1	1	1	Clean
figure_15.3	script_runs	1	1	1	Clean
figure_15.4_and_15.5	script_runs	1	1	1	Clean
figure_15.6	script_runs	1	1	1	Clean
figure_15.7	script_runs	1	1	1	Clean
figure_16.1	script_runs	1	1	1	make_dag_df left_join failure
figure_16.10	script_runs	1	1	1	Clean
figure_16.2	script_runs	1	1	1	Clean
figure_16.3	script_runs	1	1	1	Clean
figure_16.4	script_runs	1	1	1	Clean
figure_16.5	script_runs	1	1	1	Clean
figure_16.6	script_runs	1	1	1	DIDmultiplegt fails to load (rgl -> libGLU not missing on macOS Tahoe); workaround: RGL_USE_NULL=T env var; ..count.. notation deprecated in ggplot2 3.4
figure_16.7	script_runs	1	1	1	make_dag_df left_join failure
figure_16.8	script_runs	1	1	1	make_dag_df left_join failure
figure_16.9	script_runs	1	1	1	Clean
figure_17.1	script_runs	1	1	1	make_dag_df left_join failure

figure_17.2	script_runs	1	1	1	1 warning(s): Remove 30 rows containing missing values or values outside the scale range (‘geom_textpath()’).
figure_17.3	script_runs	1	1	1	Clean
figure_17.4	script_runs	1	1	1	Clean
figure_17.5	script_runs	1	1	1	Clean
figure_18.1	script_runs	1	1	1	make_dag_df left_join failure
figure_18.10	script_runs	1	1	1	Clean
figure_18.11	script_runs	1	1	1	Clean
figure_18.12	script_runs	1	1	1	vayr::sunflower() API change: width/height args renamed to density/aspect_ratio
figure_18.13	script_runs	1	1	1	Clean
figure_18.14	script_runs	1	1	1	Clean
figure_18.15	script_runs	1	1	1	Clean
figure_18.16	script_runs	1	1	1	interference package not on CRAN (install from GitHub: szonszein/interference) ggspatial also requires RGL_USE_NULL
figure_18.17	script_runs	1	1	1	1 warning(s): Remove 4 rows containing non-finite outside the scale range (‘stat_bin()’).
figure_18.2	script_runs	1	1	1	Clean
figure_18.3	script_runs	1	1	1	Clean
figure_18.4	script_runs	1	1	1	Clean
figure_18.5	script_runs	1	1	1	1 warning(s): Remove 6 rows containing missing values or values outside the scale range (‘geom_bar()’).

figure_18.6	script_runs	1	1	1	1 warning(s): Remove 8761 rows containing missing values or values outside the scale range ('geom_point()').
figure_18.7	script_runs	1	1	1	Clean
figure_18.8	script_runs	1	1	1	Clean
figure_18.9	script_runs	1	1	1	make_dag_df left_join failure
figure_19.1	script_runs	1	1	1	Clean
figure_19.2_19.3	script_runs	1	1	1	2921 warning(s): 1 observation has a computed leverage at near 1, which happens when the design is close to saturated and the observation is fitted exactly or nearly so. 'se_type = "HC2"' divides by (1 - leverage). An observation at or above leverage 1 is dropped from the variance rather than .
figure_19.4	script_runs	1	1	1	Clean
figure_19.5	script_runs	1	1	1	1 warning(s): Remove 99 rows containing missing values or values outside the scale range ('geom_segment()').
figure_19.6	script_runs	1	1	1	Clean
figure_2.1_5.1	script_runs	1	1	1	ggdag tidy_dagitty A change: select() on tidy_dagitty object returns x/y columns; left_join with nodes creates x.x/x.y conflict object 'x' not found in geom_point

figure_2.2	script_runs	1	1	1	1 warning(s): Using 'size' aesthetic for line was deprecated in ggplot2 3.4.0. Please use 'linewidth' instead
figure_21.1	script_runs	1	1	1	Clean
figure_21.2	script_runs	1	1	1	dplyr::summarise no longer allows multi-row return without refram Utility = prob return 2000 rows per group
figure_21.3	script_runs	1	1	1	Clean
figure_21.4	script_runs	1	1	1	Clean
figure_22.1	script_runs	1	1	1	Clean
figure_23.2	script_runs	1	1	1	Clean
figure_6.1	script_runs	1	1	1	Same make_dag_df left_join failure as figure_2.1_5.1
figure_6.2	script_runs	1	1	1	Same make_dag_df left_join failure
figure_8.1	script_runs	1	1	1	Same make_dag_df left_join failure
figure_8.2	script_runs	1	1	1	1 warning(s): Using 'size' aesthetic for line was deprecated in ggplot2 3.4.0. Please use 'linewidth' instead
figure_8.3	script_runs	1	1	1	Clean
figure_8.4	script_runs	1	1	1	1 warning(s): Using 'size' aesthetic for line was deprecated in ggplot2 3.4.0. Please use 'linewidth' instead
figure_8.5	script_runs	1	1	1	Clean
figure_8.6	script_runs	1	1	1	Clean
figure_8.7	script_runs	1	1	1	Same make_dag_df left_join failure

figure_9.1	script_runs	1	1	1	2 warning(s): Using 'size' aesthetic for line width was deprecated in ggplot2 3.4.0. Please use 'linewidth' instead. Removed 2 rows containing missing values or values outside the scale range ('geom_bar()').
figure_9.2	script_runs	1	1	1	Clean
figure_9.3	script_runs	1	1	1	1 warning(s): Removed 2 rows containing missing values or values outside the scale range ('geom_bar()').
figure_9.4	script_runs	1	1	1	Clean
figure_9.5	script_runs	1	1	1	Clean

Summary: 82 of 82 verifiable claims match the published values in the maintained rewrite.

6 R Environment

Table 6: R packages loaded at report compilation

Package	Version
dplyr	1.2.1
forcats	1.0.1
ggplot2	4.0.3
here	1.0.2
kableExtra	1.4.0
knitr	1.51
lubridate	1.9.5
purrr	1.2.2
readr	2.2.0
stringr	1.6.0

tibble	3.3.1
tidyr	1.3.2
tidyverse	2.0.0

R version: R version 4.6.0 (2026-04-24)

Platform: aarch64-apple-darwin23